

EcoFlux Data-Cleaning Pipeline: Stage 9 - Machine learning gap-filling in Python

The following are complete instructions on gap-filling fluxes using a machine learning method in Python. This is based on [Irvin et al. 2021](#) and has been adapted to the EcoFlux pipeline [here](#).

1. Update **Biomet**. After pulling (or downloading) the most recent changes, on the command line type **git submodule update --init --recursive**. Note: if it's your first time and you are rather cloning the repo, type
git clone --recurse-submodules https://github.com/CANFLUX/Biomet.net.git
2. Open a terminal and navigate to /Biomet.net/Python/methane-gapfill-ml on your computer. Next, run this line: **pip install -e .** (the dot or period is important!)

Note: Please make sure you have the correct Python interpreter. Sometimes it gets confused when you have more Python installed on your local computer.

3. Create a site-specific yaml file for editing titled **SiteID_ML_Gapfill.yaml** and put it here: projectFolder\Database\Calculation_Procedures\TraceAnalysis_ini\siteID\siteID_ML_Gapfill.yaml (replacing siteID with your actual siteID).
4. Any information input in the site-specific yaml will overwrite **ML_Gapfill_Default.yaml** in [Biomet.net/Python/config_files/ML_Gapfill_default.yaml](#). Every site-specific yaml must list the predictors it wants to use for training the model; otherwise, you will receive an error. Any other edits to the input trace, model type, number of splits, etc can be made in the site-specific yaml but are not necessary for the code to run. If you do not want to run gapfilling for a specific flux, input **NULL** next to the trace name. A template example of the most basic site-specific yaml file is below:

```
traces:

fch4:

  preds_trace:

    - ThirdStage/TS_F_1_1_1

    - ThirdStage/TA_F_1_1_1

    - ThirdStage/SW_IN_F_1_1_1

    - ThirdStage/SWC_1_1_1

    - ThirdStage/WS_1_1_1

nee: NULL

le: NULL

h: NULL

fno2: NULL
```

5. Train the models by running **train_ML_gapfill(yearsIn, siteID)** in MATLAB. This will run through all the ML steps, including training and gap-filling.
6. If the models have already been trained and you just want to gap-fill, use **fr_automated_cleaning(YYYY, 'siteID', 9)**

NOTE: If you make any changes to the site-specific .yml, you will need to retrain the model.

7. Outputs are written to ...projectFolder\Database\yyyy\siteID\Clean\ThirdStage_ML\ and are organized into folders by trace. Suffix descriptions are as follows:
 - a. TRACE_ML_MODEL, is the pure model output for each trace by model type (RF, ANN, XGB).
 - b. TRACE_F_ML_MODEL, is the measured trace gapfilled with the ML model
 - c. TRACE_ML_MODEL_UNCERTAINTY is the uncertainty associated with each model. See [Irvin et al. 2021](#) for details.